

A High-Precision Dual-Servo Stage Using Halbach Linear Active Magnetic Bearings

Young-Man Choi and Dae-Gab Gweon

Abstract—In this paper, a high-precision dual-servo stage with a magnetically levitated fine stage is described. For magnetic levitation, a Halbach linear active magnetic bearing (HLAMB) is used. The HLAMB has two functions: gravity compensation and actively control of vertical motion. To implement these functions, the HLAMB utilizes permanent magnet arrays and Lorentz coils. The fine stage has positioning capabilities with 6 DOF due to four HLAMBs and four voice coil motors (VCM). The HLAMBs control out-of-plane motions, and the VCM control in-plane motions. The fine stage achieves high-precision position feedback using laser interferometers and capacitive sensors. The coarse stage has an H-type structure that can carry the fine stage 300 mm by 300 mm along the x - and y -axes. Positioning and scanning performances are verified by the experimental results. It has ± 10 and ± 15 nm in-position stability in the x - and y -axes, respectively. At the scan with 10 mm/s constant velocity, ± 1 nm mean tracking error and 4.7 nm jitter were obtained.

Index Terms—Electromagnets, magnetic levitation, motion control.

I. INTRODUCTION

HIGH-PRECISION stages are widely used in the semiconductor industry, the flat panel display industry, high-precision machining, and high-precision measuring machines. These industries demand high precision and high throughput. Dual-servo stages have been developed to achieve high precision and a fast response time over large strokes [1]–[7]. A dual-servo stage has two components: a coarse stage and a fine stage. The fine stage, connected to the coarse stage in series or in parallel, improves positioning accuracy, resolution, and bandwidth, whereas the coarse stage carries the fine stage over long strokes. Wafer stages, which present the greatest challenges for stage technology development, have been developed to integrate a dual-servo stage with a magnetically levitated (maglev) fine stage [1]–[4]. Magnetic levitation removes any mechanical contacts that may degrade high-precision positioning and scanning. Because the maglev stage is able to move with 6 DOF, it can compensate for all spatial components of motion error. Magnetic levitation yields superior vibration isolation than conventional mechanical guide systems, which are characterized by

a high stiffness (10^7 – 10^8 N/m) in the vertical direction, and offer poor isolation from low-frequency ground vibrations. However, a magnetic levitation stage can attenuate frame vibrations by actively controlling the transmissibility. In addition, the quasi-static stiffness of a maglev stage can theoretically be infinite by servo control (especially integral control). This characteristic has benefits over the limited static stiffness of a passive bearing system, especially when quasi-static process forces and/or disturbance-related forces are present, as occur in imprint lithography or hot-embossing applications [8]. On the other hand, there are some difficulties associated with implementing magnetic levitation. By compensating for gravity, such systems consume power and inevitably generate heat that could change the properties of actuators or sensors and deform mechanical structures, even specimens. To reduce power consumption, many maglev systems have adopted solenoids because of their large force capabilities. However, solenoid-based actuators require nonlinear control, such as feedback linearization, to linearize their nonlinear behavior, which increases the complexity of control [9], [10]. Last, a magnetic levitation stage requires simultaneous 6 DOF control, which entails a huge computational burden [11].

As a key component of the maglev stage, our group previously proposed a Halbach linear active magnetic bearing (HLAMB) that has very small power consumption by exactly compensating for gravity [12]. The gravity-compensating force is generated by permanent magnets. Furthermore, a HLAMB actively controls a target without feedback linearization due to its linear force-to-current relation. The most prominent advantage of a HLAMB is that it has a negligible vertical stiffness. In this paper, we developed a 6 DOF magnetically levitated stage using the HLAMBs. Moreover, a high-precision long-stroke dual-servo stage is developed. Fig. 1 shows the proposed dual-servo stage including the maglev fine stage. A coarse stage delivers a fine stage over long strokes in the x - and y -directions. On the coarse stage, the fine stage is magnetically levitated by four HLAMBs. As well as compensating for the gravity of the moving platform, they actively control the fine stage over 3 DOF: the z , θ_x , and θ_y . Four voice coil motors (VCM) are used to control the in-plane motion; the x -, y -, and θ_z -axes. All of the actuators are symmetrically located to implement a simple 6 DOF control scheme. This paper is organized as follows. In Section II, the principles underlying the HLAMB are briefly introduced. In Section III, concept designs for the 6 DOF maglev and coarse stages are presented. In Section IV, the 6 DOF control strategy for the maglev stage is presented using the rigid body modeling and decoupling method. In Section V, experimental results for high-precision positioning and scanning are given. Finally, Section VI provides conclusions on this paper.

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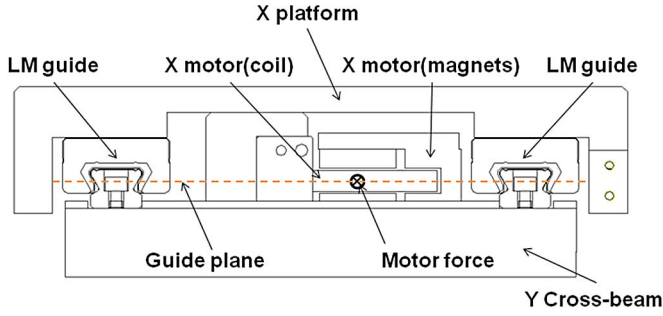


Fig. 4. X platform of coarse stage.

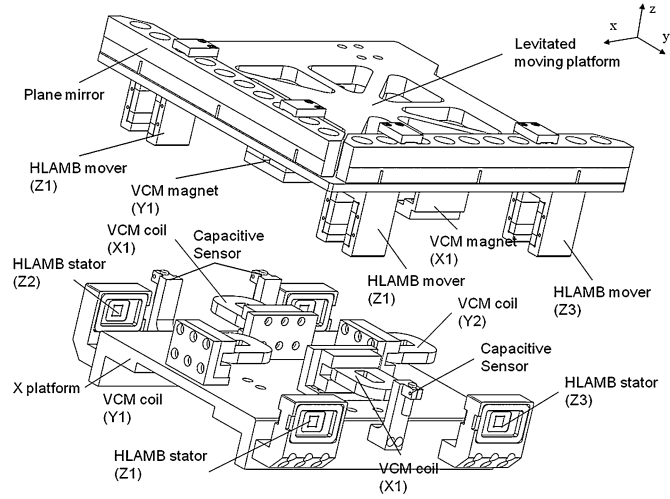


Fig. 5. Six DOF maglev fine stage (exploded view).

of the X platform is also measured by a linear scale (not shown here). The Y crossbeam and the X platform are machined from aluminum 6061-T6.

B. Design of the Maglev Fine Stage

The fine stage is designed to satisfy the following principles for high-precision motion.

1) *Direct Drive*: Direct drive in all axes guarantees no wear and no friction. In particular, moving-magnet direct drive can eliminate wiring connections to the moving platform. The fine stage utilizes moving-magnet VCM as well as the HLAMB.

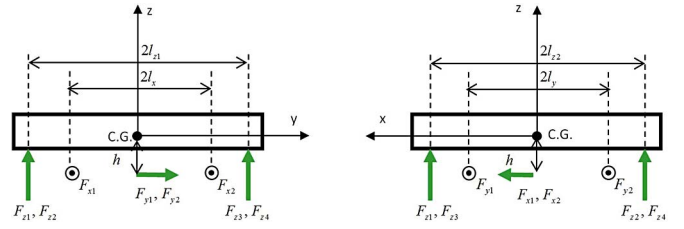
2) *Linear Characteristics*: As previously discussed, linear characteristics of an actuator can reduce the control complexity and cost. Both HLAMB and VCM have good linear characteristics.

3) *Symmetric Actuation*: All actuators are placed symmetrically. Four HLAMBs are used for out-of-plane motion and four VCM are used for in-plane motion. The presence of an additional redundant actuator produces simpler kinematics, and the system consumes less power than the minimum actuation case.

Fig. 5 shows the proposed magnetic levitation stage. Four HLAMBs are symmetrically placed at each corner of the moving platform. Each HLAMB compensates one-fourth of the gravity and simultaneously controls 3 DOF, which are the z -, θ_x -, and θ_y -axes. For in-plane motion along the x -, y -, and θ_z -axes,

TABLE I
PARAMETERS OF THE DUAL-SERVO STAGE

	Parameters	Value	Unit
Coarse stage	moving mass in x-axis	6.5	kg
	moving mass in y-axis	33.5	kg
	Stroke in x-axis	300	mm
	Stroke in y-axis	300	mm
	max. acceleration in x-axis	20	m/s ²
	max. acceleration in y-axis	8.3	m/s ²
Fine stage	moving mass	9.75	kg
	max. force in x- and y-axes	238	N
	max. force in z-axis	52.2	N
	working volume	1×1×1	mm ³

Fig. 6. Actuator locations of fine stage. (Left) yz -plane. (Right) xz -plane.

four VCM are utilized because it has a linear current-to-force relation without any mechanical coupling. We designed a VCM to operate with a working gap that is necessary for levitation. Each VCM generates 14.8 N/A. In Fig. 5, front and back VCM contribute to the x and θ_z motion, whereas left and right VCM contribute to the y and θ_z motion. All coil parts in the HLAMBs and VCM are placed on the X platform, whereas the movers, which contain only permanent magnets, are fixed to the moving platform of the fine stage. This moving magnet configuration prevents heat, which is generated by the coils, from being transferred to a target area. This configuration also removes the wiring connections from the coarse stage to the moving platform. The fine stage has a 1 mm × 1 mm × 1 mm travel range by itself.

For 6 DOF positional measurements, three laser interferometers and three capacitive sensors are used. The moving platform of the fine stage has two bar-shaped plane mirrors, which are orthogonally placed. Laser interferometers measure in-plane positions using the mirrors: two interferometers measure x -axis position and θ_z -axis position, and the other measures the y -axis position. To maintain orthogonality, each plane mirror is fixed using positioning pins and two fixtures. However, the limited stiffness of the fixtures introduces a rocking mode into the system. To damp this mode, a viscoelastic sheet is inserted between the fixtures and the mirror. Beneath the moving platform, three capacitive sensors are installed to measure vertical displacements. Parameters of the developed dual-servo stage are summarized in Table I.

IV. CONTROL STRATEGY FOR THE DUAL-SERVO STAGE

As shown in Fig. 6, the fine stage has eight force sources: F_{x1} , F_{x2} , F_{y1} , and F_{y2} from the VCM, and F_{z1} , F_{z2} , F_{z3} , and F_{z4} from the HLAMBs. Each force contributes a coupled motion of

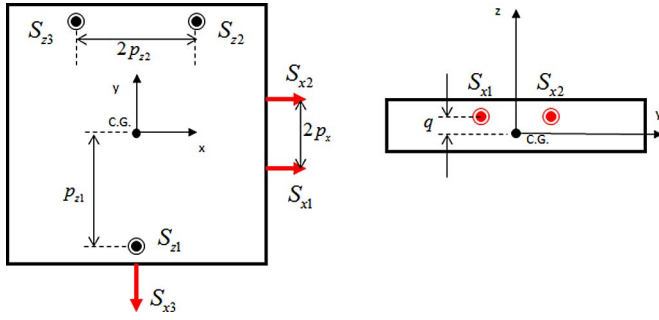


Fig. 7. Sensor locations of fine stage. (Left) xy -plane. (Right) yz -plane.

the motion platform. Therefore, the proposed system is defined as a multi-input–multi-output (MIMO) system. To implement a practical control for the MIMO system, a pseudoinverse method [13] was used.

First, a Cartesian coordinate frame is defined at the initial position of the center of gravity of the moving platform. This absolute coordinate system is fixed at the base. Then, a rigid body mode is modeled assuming that the HLAMBs and VCMs have zero stiffness and damping. Because the moving platform undergoes extremely small rotations, the effect of rotation on the forces is ignored. The off-diagonal terms in the moment of inertia are also assumed to be zero. In this case, the coarse and fine stages are both modeled as double integrators. Thus, the simplified rigid body model for the fine stage is as follows:

$$\mathbf{M}\ddot{\mathbf{x}} = \mathbf{F}_{\text{kin}}\mathbf{f} \quad (1)$$

$$\mathbf{M} = \text{dia}[M_f \ M_f \ M_f \ (I_{xx})_f \ (I_{yy})_f \ (I_{zz})_f] \quad (2)$$

$$\mathbf{F}_{\text{kin}} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & h & h & -l_{z1} & -l_{z1} & l_{z1} & l_{z1} \\ -h & -h & 0 & 0 & -l_{z2} & l_{z2} & -l_{z2} & l_{z2} \\ l_x & -l_x & l_y & -l_y & 0 & 0 & 0 & 0 \end{bmatrix} \quad (3)$$

where $\mathbf{M}$ is the mass matrix in which M_f is the mass of the moving platform, $(I_{mm})_f$ is the moment of inertia of the m -axis, and $\mathbf{x}$ is the position of the fine stage, whose components are $\{x, y, z, \theta_x, \theta_y, \theta_z\}$, $\mathbf{f}$ is the forces from VCM and HLAMBs, whose components are $\{F_{x1}, F_{x2}, F_{y1}, F_{y2}, F_{z1}, F_{z2}, F_{z3}, F_{z4}\}$ and $\mathbf{F}_{\text{kin}}$ is a forward kinematic matrix in which h is the distance from the force line of the VCM to the center of gravity, l_x is half of the distance between two X VCM, l_y is half of the distance between two Y VCM, l_{z1} is half of the distance between HLAMBs in the yz -plane, and l_{z2} is half of the distance between HLAMBs in the xz -plane.

Next, the sensor coordinates are transformed to determine the position of the center of gravity. Six sensors independently measure the corresponding positions of the moving platform. Fig. 7 shows the locations of sensors and the center of gravity.

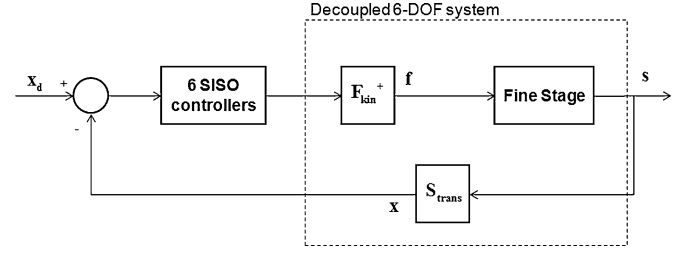


Fig. 8. Control block diagram for the fine stage.

Equation (4) shows the transformation from the sensor measurements $\mathbf{s} = \{S_{x1}, S_{x2}, S_y, S_{z1}, S_{z2}, S_{z3}\}$ to the absolute position of the center of gravity.

$$\mathbf{x} = \mathbf{S}_{\text{trans}}\mathbf{s} \quad (4)$$

$$\mathbf{S}_{\text{trans}} = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 & \frac{q}{2p_{z2}} & \frac{-q}{2p_{z2}} \\ 0 & 0 & 1 & \frac{q}{2p_{z1}} & \frac{-q}{4p_{z1}} & \frac{-q}{4p_{z1}} \\ 0 & 0 & 0 & 0.5 & 0.25 & 0.25 \\ 0 & 0 & 0 & \frac{-1}{2p_{z1}} & \frac{1}{4p_{z1}} & \frac{1}{4p_{z1}} \\ 0 & 0 & 0 & 0 & \frac{-1}{2p_{z2}} & \frac{1}{2p_{z2}} \\ \frac{-1}{2p_x} & \frac{1}{2p_x} & 0 & 0 & 0 & 0 \end{bmatrix} \quad (5)$$

Fig. 8 presents a control block diagram using the pseudoinverse method. The matrix $\mathbf{F}_{\text{kin}}^+$ is a pseudoinverse of the forward kinematics matrix (3), which converts the forces acting on the center of gravity into the actuator forces. The pseudoinverse is calculated by the Moore–Penrose matrix inverse, which is given as (6).

$$\mathbf{F}_{\text{kin}}^+ = (\mathbf{F}_{\text{kin}}^H \mathbf{F}_{\text{kin}})^{-1} \mathbf{F}_{\text{kin}}^H \quad (6)$$

where the superscript $\mathbf{H}$ means the conjugate-transpose matrix.

Using the transformation matrix, $\mathbf{S}_{\text{trans}}$ in (5), the position of the center of gravity can be determined from six sensor measurements. In the view of the controller, the MIMO system can be regarded as six independent single-input–single-output (SISO) systems, and each axis can be controlled independently as a SISO system. This method is reasonable because the decoupling error caused by the pseudoinverse is extremely small.

Since the fine stage is stacked on the coarse stage, the fine stage imparts reaction forces on the coarse stage. The model for the coarse stage is given as follows:

$$M_{cx}\ddot{x}_c = F_{cx} - F_{x1} - F_{x2} \quad (7)$$

$$(M_{cy} + M_{cx})\ddot{y}_c - M_{cx}x\ddot{\theta}_c = F_{cy1} + F_{cy2} - F_{y1} - F_{y2} \quad (8)$$

$$(I_{c1} + I_{c2} + M_{cx}x^2)\ddot{\theta}_c - M_{cx}x\ddot{y}_c = l_c(F_{cy2} - F_{cy1}) - l_x(F_{x1} - F_{x2}) - l_y(F_{y1} - F_{y2}) \quad (9)$$

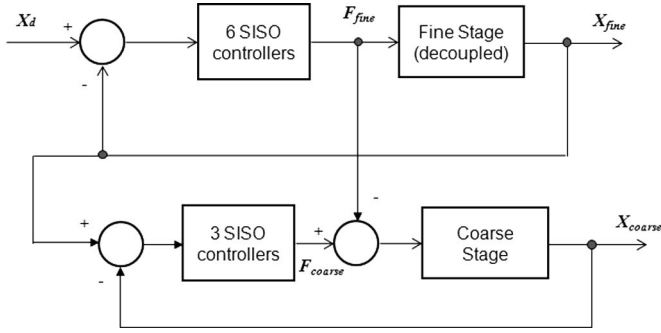


Fig. 9. Control block diagram for the dual stage.

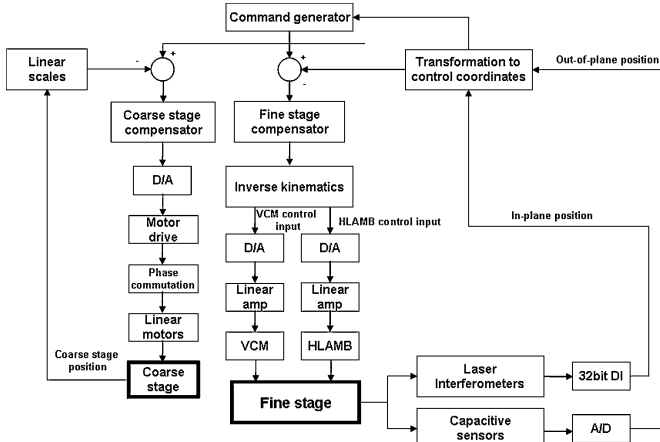


Fig. 10. High-level control block diagram for the total dual-servo system.

where M_{cx} is the mass of the X platform, M_{cy} is the mass of the Y cross beam, I_{c1} is the z-axis moment of inertia of the X platform, I_{c2} is the z-axis moment of inertia of the Y cross beam, F_{cx} , F_{cy1} , F_{cy2} are the actuating forces of the coarse stage, and l_c is the distance between F_{cy1} and F_{cy2} . Considering the reaction forces from the fine stage, the coarse stage should be designed to generate enough forces, which can compensate the reaction in addition to its own thrust force.

The dual-servo system is controlled by the master–slave dual-servo control scheme with two sensors [14]. Fig. 9 shows the scheme that the coarse stage follows the measured position of the fine stage. Conventionally, the fine stage compensates for the following error of the coarse stage. However, when the system has two individual sensors, one for each stage’s servo, and especially, when the fine stage’s sensor is more accurate than that of the coarse stage, the conventional scheme could lead to the degradation of the positioning accuracy of the fine stage. In this case, it is better that the coarse stage follows the fine stage, just keeping a sufficient moving margin between the two stages. The fine stage tracks the desired command with high-precision feedback and high bandwidth.

The overall control architecture is shown in Fig. 10. The command generator makes the desired position commands to the coarse stage and the fine stage. Each compensator calculates control inputs from the difference between the command and the transformed position. By inverse kinematics, the control

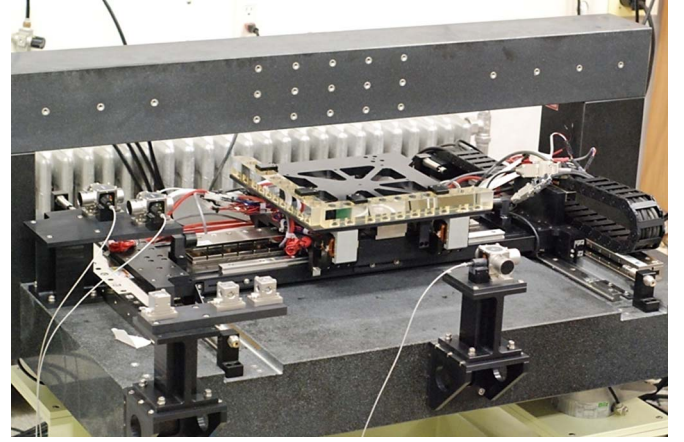


Fig. 11. Picture of the total system including the measurement system.

inputs are converted to actuator forces. Linear amplifiers supply a current corresponding to the calculated actuator force. Position data, obtained by laser interferometers and capacitive sensors, are transformed to the absolute coordinates of the center of gravity.

V. EXPERIMENTAL RESULTS

In this section, we present the results of 6 DOF control experiments demonstrating minimum resolutions and dynamic performance of the system. Fig. 11 shows the total system setup. The dual-servo stage and all measurement systems are installed on a granite base, which is suspended by four pneumatic vibration isolators. The total footprint of the system is 1440 mm × 1145 mm × 1120 mm. The real-time controller, based on a DSP (DS1005, dSPACE), manages the hardware and calculates the control outputs. This controller has 12 channels of 16-bit D/A outputs and six channels of analog encoder interface with a 10-kHz sampling frequency. To drive the coreless permanent magnet linear motors (LZ-030, Anorad) in the coarse stage, we use linear amplifiers (TA320, Trust Automation), with the commutation function, using a hall sensor. As feedback sensors for the coarse stage, linear scales (RGH24, Renishaw) are used, which have an analog output with a 20- μ m pitch that gives a resolution of 4.88 nm after digital interpolation. For the HLAMBs and VCM, dc linear current amplifiers (TA115, Trust Automation) are utilized, because they give higher positioning stability than pulsewidth modulation (PWM) amplifiers. The in-plane components of the position of the fine stage are measured by plane mirror interferometers (10706 A, Agilent) with a resolution of 4.95 nm. The out-of-plane components are measured by three capacitive sensors (2800 series, ADE) with a gain of 2.5 μ m/V. The dual-servo control scheme is programmed into the controller as well as the PID controller for each axis. The bandwidth of the x- and y-axis is about 300 Hz, while that of the z-axis is about 30 Hz.

Levitation is accomplished through the following steps: first, the fine stage is moved to a mechanical home position. The in-plane control is started. To secure a moving margin, the fine stage is moved to the center of the workspace. After stabilization for several seconds, out-of-plane control, using HLAMB and

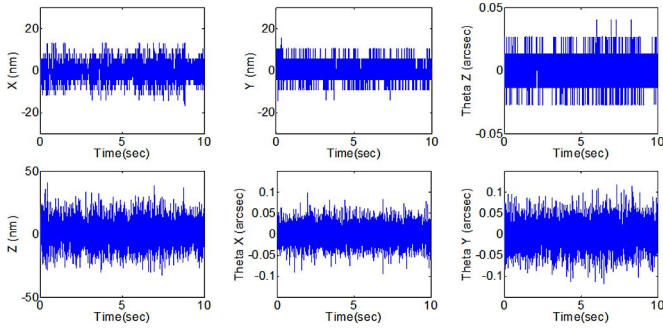


Fig. 12. In-position stabilities.

TABLE II
IN-POSITION STABILITIES

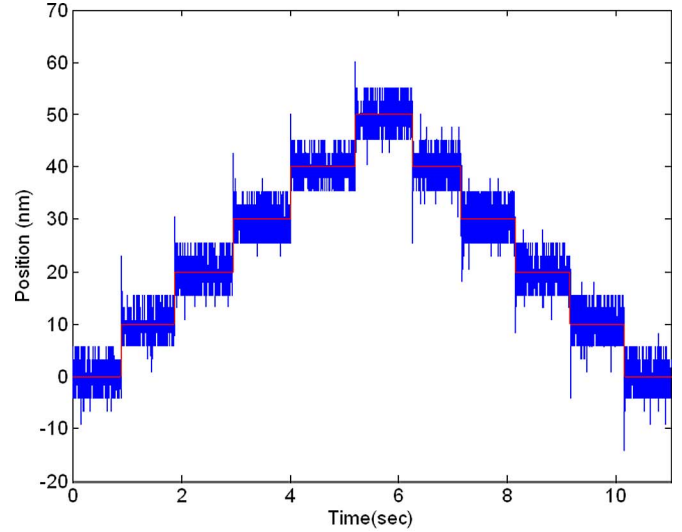
	axis	Peak-to-peak	rms
In-plane stabilities	x	19.8 nm	3.1 nm
	y	29.6 nm	3.9 nm
	θ_z	0.07 arcsec	0.01 arcsec
Out-of-plane stabilities	Z	73.1 nm	9.6 nm
	θ_x	0.19 arcsec	0.02 arcsec
	θ_y	0.24 arcsec	0.03 arcsec

capacitive sensors, is started. In the last step of the process, the maglev stage is slowly levitated to $5 \mu\text{m}$. After stabilizing, the dual-servo system is ready for motions.

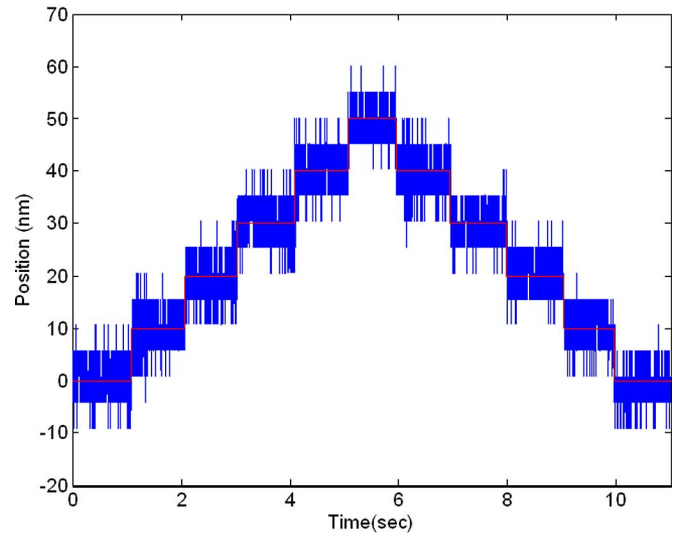
Fig. 12 shows the in-position stability during levitation for the 6 DOF. The in-plane positions were kept within ± 2 to ± 3 counts of the laser interferometer signal. The stability in the x -direction was shown to be better than that in the y -direction because the x -axis position is averaged using two laser interferometer signals. The stabilities of the out-of-plane axes are worse than that of the in-plane axes. Because the capacitive sensor uses an analog voltage output, electrical noise in the output degrades the resolution. Moreover, the servo bandwidth cannot be increased, because it contains high-frequency components. Thus, the servo bandwidth for out-of-plane motion is lower than the bandwidth for in-plane motion. Table II shows the peak-to-peak and rms values of in-position stabilities for all axes.

Power consumption in levitation was estimated by the current during levitation maintaining a position. The control inputs of the four HLAMBs were 0.16, -0.08 , 0.38, and 0.14. These offset currents contributed toward compensating the mass mismatch between the ideal and actual masses, and the asymmetrical moment of inertia. In this case, 670 mW was consumed for levitation, which hardly affects the domains of interest in terms of thermal stress.

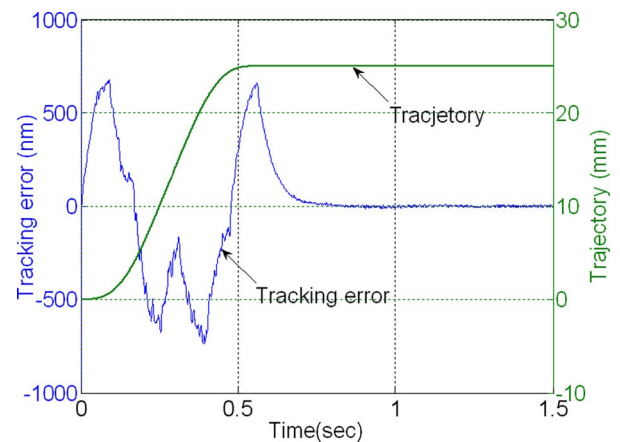
To assess the resolution of the proposed stage, a series of 10 nm steps was tested in the x - and y -directions. The 10-nm steps are clearly resolved in both directions, as shown in Fig. 13. Considering the minimum sensor resolution is about 5 nm, this stage can achieve better precision if sensors with finer resolution are used. Next, a large step motion was also tested. A 25-mm step motion was implemented with a smooth trajectory of sinusoidal functions. The maximum acceleration is 0.5 m/s^2 . Fig. 14 shows the position error during the step motion. After the end of the trajectory, it took 88 ms to settle within a boundary of 100 nm.



(a)



(b)

Fig. 13. Minimum resolution test showing 10 nm resolution. (a) x -axis. (b) y -axis.Fig. 14. Position error with 25 mm step in the x -axis.

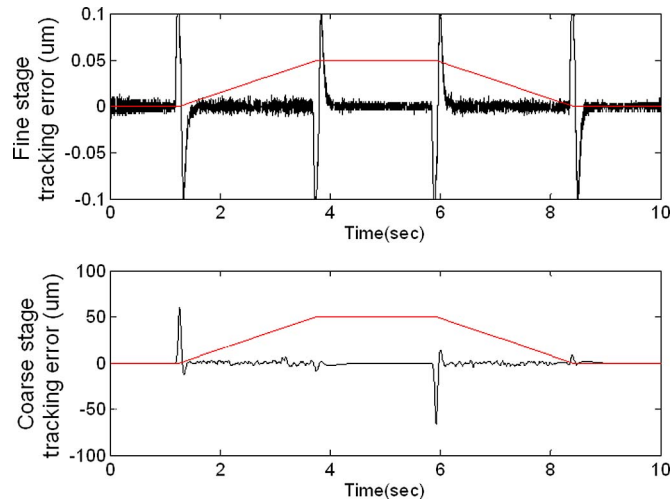


Fig. 15. Tracking errors of the fine stage and the coarse stage for a 10 mm/s scan.

Lastly, a high-precision scanning performance, one of the most important requirements for high-precision stages, was evaluated. Fig. 15 shows the tracking errors of the fine stage and the coarse stage for a scan trajectory with 10 mm/s. Note the scales of the vertical axes in two graphs. The fine stage shows great scanning performance, which exhibits almost 500 times smaller tracking error than the coarse stage. The averaged tracking error during scanning was 3.0 nm and the rms error was 3.7 nm. In [2], they developed a wafer stage using magnetic levitation for extreme UV lithography. In this reference, scanning performance was evaluated using two parameters: 1) the mean tracking error, defined as the averaged mean of the tracking error measured during motion at a constant velocity over a 150 ms moving window average; and 2) jitter, defined as the rms of the tracking error measured during motion at a constant velocity over a 150 ms moving window average. The mean tracking error and jitter with 10 mm/s scan were ± 1.5 and 4.5 nm, respectively. In our case, the mean tracking error is less than ± 0.97 nm and the jitter is less than 4.7 nm. In terms of mean tracking error, our stage outperforms the wafer stage in [2]. Considering that the control system is not optimized yet, the scan performance could be improved even more.

VI. CONCLUSION

We developed a high-precision dual-servo stage. The coarse stage was designed as an H-type XY stage. The fine stage is magnetically levitated to achieve 6 DOF positioning capabilities. The fine stage has four HLAMBs and four VCM. The HLAMBs control 3 DOF (z , θ_x , and θ_y) and the VCM control the other 3 DOF (x , y , and θ_z). The basic performance characteristics of the proposed system were evaluated through a series of experiments. The positioning resolution in the x - and y -directions was 10 nm, and the rms in-position stabilities in the x -, y -, and z -directions were smaller than 4 and 10 nm, respectively. Also, scanning performance was assessed to verify the high-precision tracking capability. These results demonstrate that the proposed dual-servo stage is suitable for a next generation high-precision stage.

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